



Smart Tourism Automation Platform using AI & n8n

AI /ML

AI & AUTOMATION

TRAVEL TECHNOLOGY



Project Overview

Smart Tourism Automation Platform is an AI-powered automation system built using n8n to simplify common travel-related tasks.

What the system does

- Collects customer travel preferences
- Generates personalized travel recommendations
- Manages booking and itinerary information
- Sends weather-based travel alerts
- Analyzes customer feedback using AI

Core Idea

Connect AI + Automation + Travel Services into one streamlined workflow.

Platform: n8n | AI: OpenAI | Services: Google Sheets, Gmail, Google Calendar & Weather API

Introduction

→ What is the Smart Tourism Automation Platform?

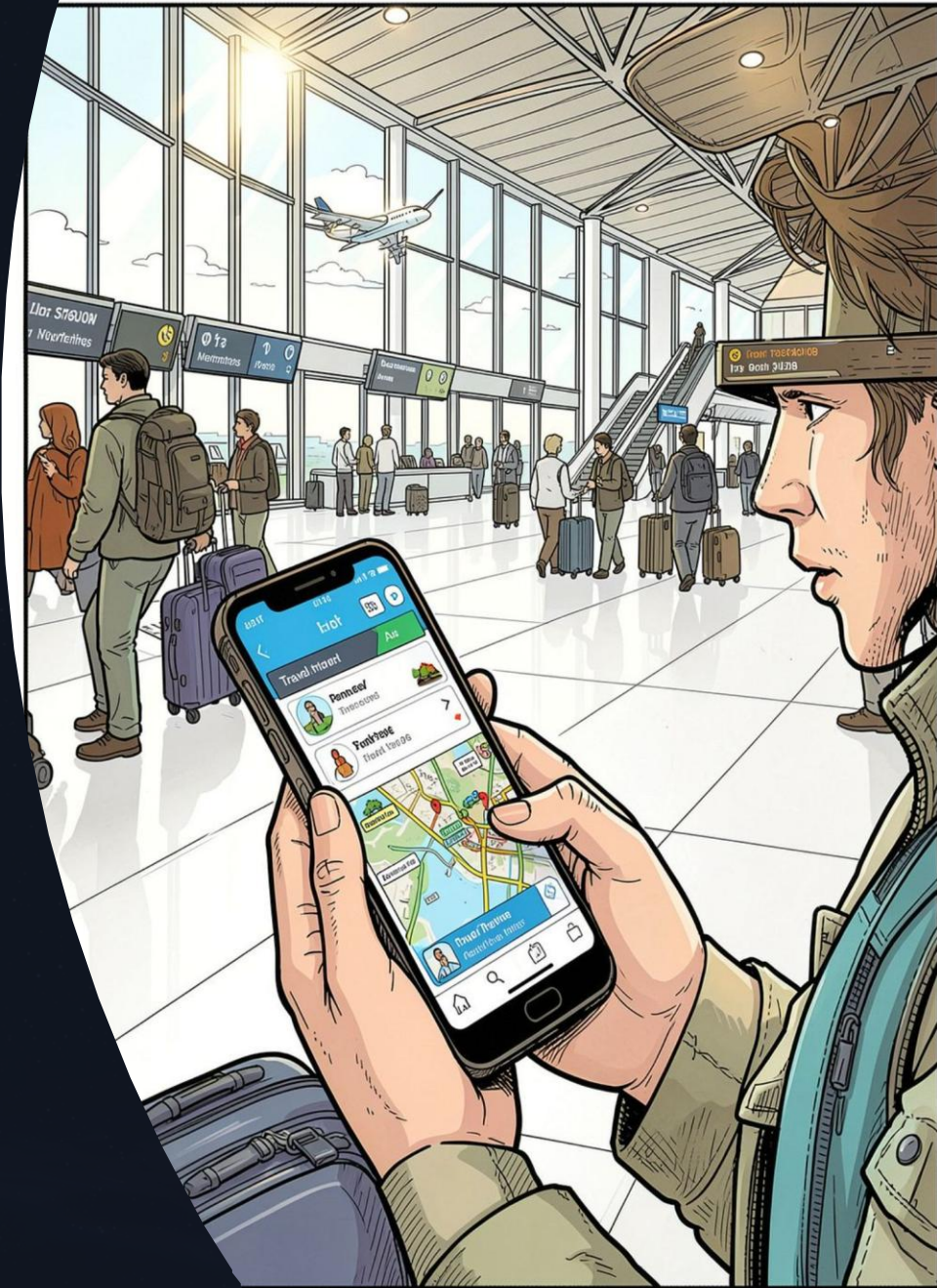
An intelligent system that automates end-to-end travel planning — from registration and AI-powered recommendations to booking, itinerary generation, real-time alerts, and post-trip feedback collection.

→ Why Travel Automation is Useful

Manual travel planning is time-consuming and error-prone. Automation reduces human effort, ensures consistency, delivers personalised experiences, and keeps travellers informed at every stage of their journey.

→ Core Idea

Combine AI (OpenAI) for smart decision-making with n8n for workflow orchestration — creating a seamless, automated tourism experience.



Problem Statement & Objectives

Travel Planning Pain Points

- Overwhelming amount of information to research manually
- No personalised recommendations based on preferences
- Fragmented tools for booking, scheduling, and alerts
- Lack of real-time travel updates and weather notifications
- No structured feedback loop to improve future trips

Main Objectives of the Project

- Automate customer registration and data capture via webhooks
- Generate AI-powered personalised travel recommendations
- Auto-create itineraries and send booking confirmations
- Deliver real-time travel alerts (weather, reminders)
- Collect and analyse post-trip feedback automatically

Proposed Solution

How AI + n8n automates travel planning, booking, alerts and feedback.



Customer Registration

Webhook captures user preferences, destination interest, budget, and travel dates automatically into Google Sheets.



AI Recommendation

OpenAI Agent analyses user data and suggests personalised destinations, activities, and travel tips.



Booking & Itinerary

n8n auto-generates a day-wise itinerary, sends Gmail confirmations, and creates Google Calendar events.



Travel Alerts

Weather API triggers automated alerts for adverse conditions; reminder emails are sent before departure.



Feedback & Analytics

Post-trip feedback forms are sent via Gmail; responses are logged in Google Sheets for analytics.

Detailed System Workflow

The unified architecture covering all 5 workflows with key integrations: Webhooks, AI Agent/OpenAI, Google Sheets, Gmail, Google Calendar, and Weather API.



The diagram above illustrates the end-to-end automated flow — from initial user registration through AI-powered recommendations, booking and itinerary creation, real-time travel alerts, and finally post-trip feedback collection and analytics — all orchestrated by n8n with integrated services.

AI & Automation



AI Agent (OpenAI)

The OpenAI Agent acts as the intelligent core — it reads user preferences from Google Sheets, understands natural language queries, and generates context-aware travel recommendations, itinerary descriptions, and personalised tips.



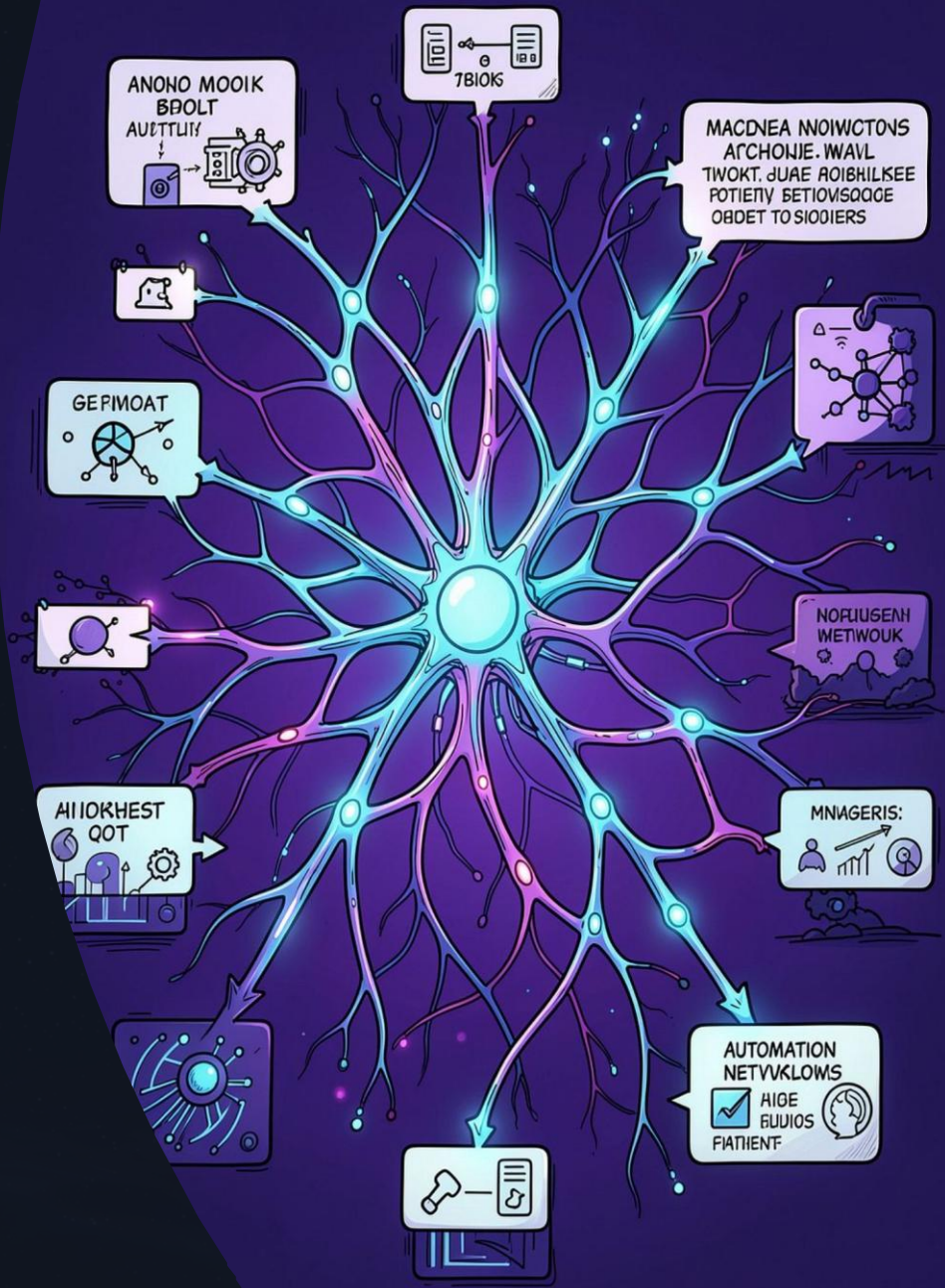
n8n Automation

n8n orchestrates all workflows without manual intervention. It connects webhooks, AI agents, APIs, and Google services into seamless automated pipelines — triggering actions based on events and conditions.



Event-Driven Triggers

Every workflow is triggered by specific events — a new form submission, a scheduled time, or an API response — ensuring the system reacts in real time without human input.



Technologies & Integrations



n8n

Open-source workflow automation platform that connects all services and orchestrates the 5 core workflows.



Webhooks

Capture real-time user inputs from registration forms and trigger downstream automation workflows instantly.



OpenAI

Provides the AI Agent for generating personalised travel recommendations, itineraries, and smart suggestions.



Google Sheets

Stores user registration data, preferences, booking details, and feedback responses in structured tables.



Gmail

Sends automated booking confirmations, travel alerts, reminders, and post-trip feedback request emails.



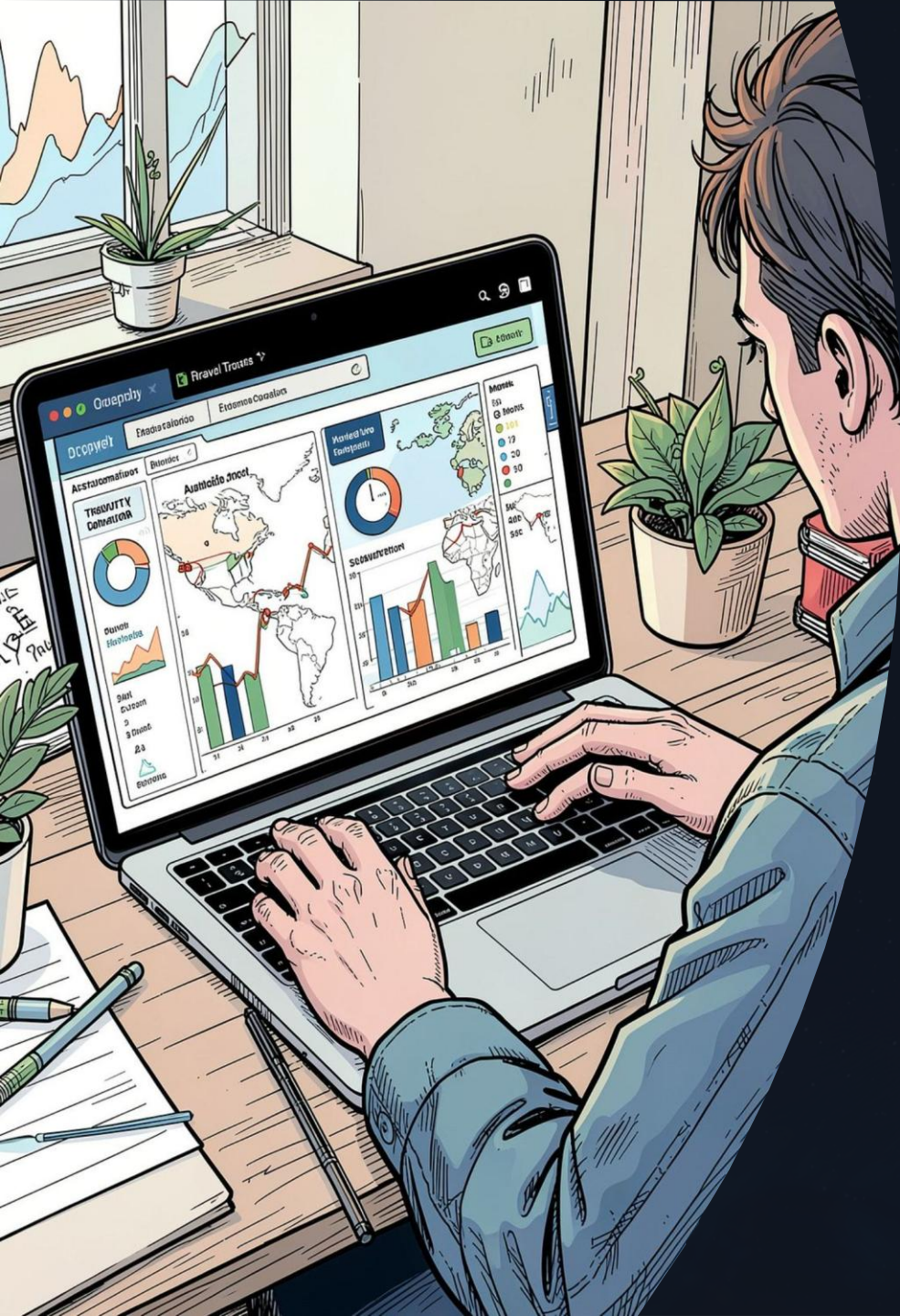
Google Calendar

Auto-creates calendar events for each day of the itinerary, keeping travellers organised and on schedule.



Weather API

Fetches real-time weather data for destinations and triggers automated alerts for adverse conditions.

An illustration of a person with dark hair, seen from the side, sitting at a desk and working on a laptop. The laptop screen displays a dashboard with several data visualizations: a world map with a red line, a bar chart, a line graph, and a pie chart. The dashboard has a header with 'Ocepey' and 'Travel Trends'. The person's hands are on the laptop keyboard. On the desk, there is a small potted plant and some papers. The background shows a window with a view of a landscape.

Implementation & Results

1 What Was Implemented

All 5 workflows were built and tested in n8n — Customer Registration (webhook → Google Sheets), AI Recommendation (OpenAI Agent), Booking & Itinerary (Gmail + Google Calendar), Travel Alerts (Weather API + Gmail), and Feedback & Analytics (Gmail + Sheets).

2 Main Outputs & Results

The system successfully auto-generated personalised itineraries, sent booking confirmation emails, created calendar events, delivered weather-based alerts, and collected structured feedback — all without manual steps.

3 Key Achievement

Demonstrated a fully automated, end-to-end smart tourism pipeline using free-tier tools — proving that intelligent travel automation is achievable with accessible technology.

Future Scope



Live Booking APIs

Integrate real flight, hotel, and transport booking APIs (Skyscanner, Amadeus) for end-to-end reservations.



More Travel APIs

Add APIs for attractions, local transport, currency conversion, and visa information for richer recommendations.



Better Analytics

Build a visual analytics dashboard to track user behaviour, popular destinations, and feedback trends.



Error Handling / Retry

Implement robust error handling and automatic retry logic in n8n workflows for production reliability.



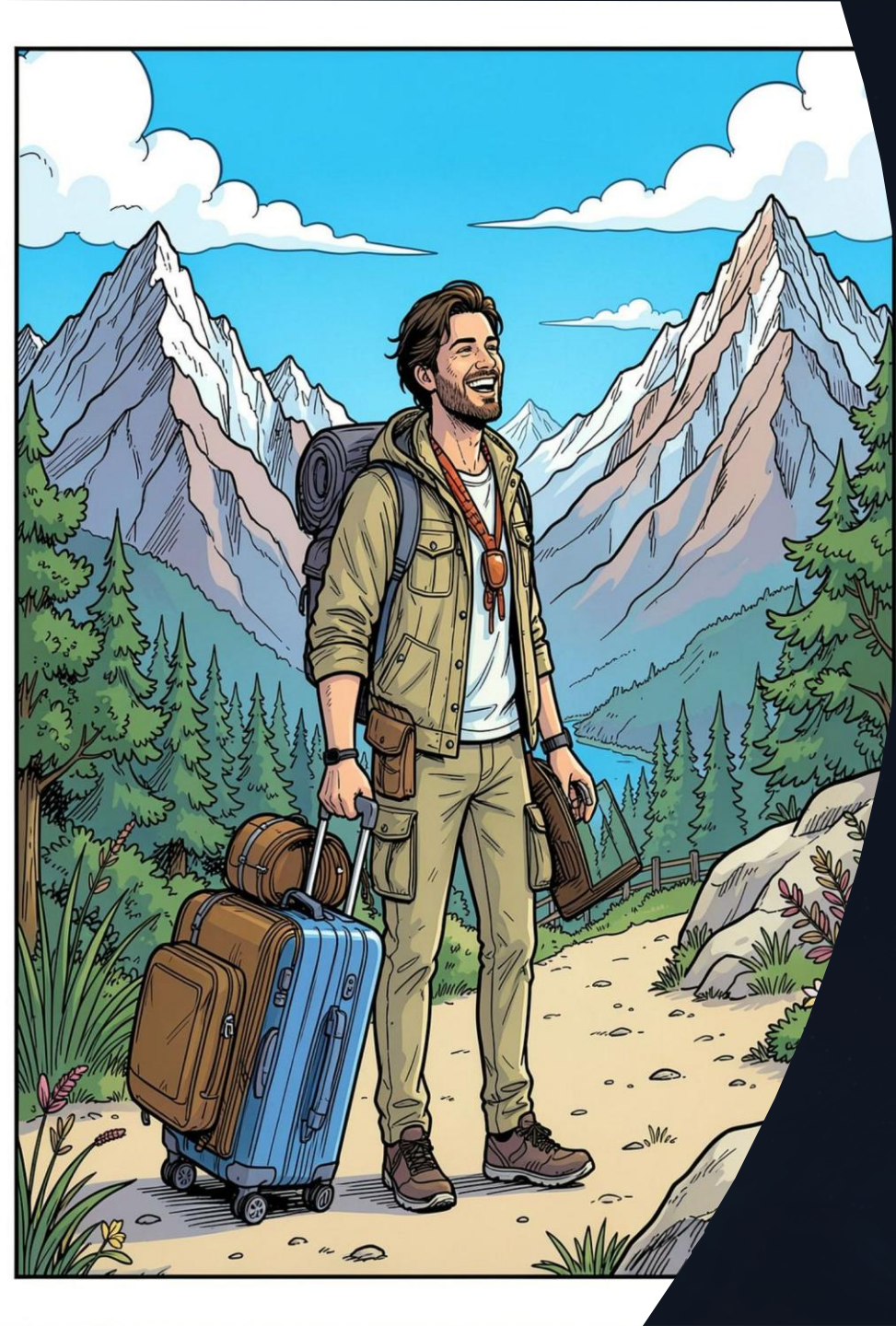
Human Approval

Add a manual approval step before final booking confirmation for user control and safety.



More Destinations

Expand the AI knowledge base to cover more global destinations, languages, and cultural contexts.

A colorful illustration of a man with a beard and long hair, smiling as he looks towards a mountain range. He is wearing a tan jacket, a white t-shirt, and tan pants. He has a large backpack and is pulling a blue rolling suitcase. The background features snow-capped mountains, green pine trees, and a clear blue sky with white clouds.

Conclusion

Project Summary

The Smart Tourism Automation Platform successfully demonstrates how AI (OpenAI) and workflow automation (n8n) can transform the travel planning experience — making it faster, smarter, and fully automated.

Key Benefits

- Eliminates manual effort in travel planning and booking
- Delivers personalised, AI-driven recommendations
- Keeps travellers informed with real-time alerts
- Creates a closed feedback loop for continuous improvement

Overall Outcome

This project proves that a fully automated, intelligent tourism platform is achievable using accessible, open-source tools — paving the way for smarter, more efficient travel experiences in the future.